Chapter 4. Linear Structural Causal Models

Path coefficients, Wright's rules, single-door and instrumental criteria

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Motivation. When can a structural coefficient of a linear Gaussian SCM be computed from the covariance matrix, and when does regression return it?

Outline:

(1) linear SCMs and Wright's path-tracing rules;

(2) when regression finds causal effects (single-door, back-door);

(3) algorithmic identification (polynomial systems, IVs, conditional IVs, instrumental sets); appendices on complexity, reversed regression and Gröbner bases.

4.1 Linear structural causal models

Motivation. Specialize SCMs to linear mechanisms and read covariances off the graph.

4.1.1 Definition and conventions

A linear SCM writes each variable as a linear function of its direct causes,

$$Y = \sum_i \lambda_{x_i y} X_i + \mathcal{E}_y,$$

with $\lambda_{x_i y}$ the change in Y per unit change of X_i and all error covariances specified. WLOG $\mathbb{E}[X] = 0$, $\mathbb{E}[X^2] = 1$ and $\mathcal{E} \sim \mathcal{N}$, so P is fixed by $\Sigma = (\sigma_{ij})$. Only f_i changes from the non-parametric case:

$$V_i \leftarrow f_i(\mathbf{pa}_i, \mathbf{u}_i) \quad \rightsquigarrow \quad V_i \leftarrow \sum_{V_j \in Pa_i} \lambda_{ji} V_j + \mathcal{E}_i.$$

(1) λ_{ji} is the *structural coefficient*;

(2) U_i is renamed $\mathcal{E}_i$;

(3) all λ_{ji} give all causal effects.

Example 4.1

$X \leftarrow f_X(U_X)$, $W \leftarrow f_W(U_W)$, $Y \leftarrow f_Y(X, W, U_Y)$ becomes $X \leftarrow \mathcal{E}_x$, $W \leftarrow \mathcal{E}_w$, $Y \leftarrow \lambda_{xy} X + \lambda_{wy} W + \mathcal{E}_y$; coefficients on the edges specify the model.

Latent confounding. $\epsilon_{ij} = \mathbb{E}[\mathcal{E}_i \mathcal{E}_j]$ labels a bidirected edge $V_i \leftrightarrow V_j$; here it means *error covariance*, not necessarily a shared latent.

Interventions. For $X \xrightarrow{\lambda} Y$,

$$\mathbb{E}[Y \mid \mathrm{do}(X = x)] = \mathbb{E}[\lambda x + \mathcal{E}_y] = \lambda x + \mathbb{E}[\mathcal{E}_y] = \lambda x.$$

4.1.2 The identification problem

Given (i) a graph $\mathcal{G}$ and (ii) data $\{(x^{(i)}, y^{(i)})\}_{i=1}^N$ on all observed variables, but (iii) neither the λ (the targets) nor the ϵ.

4.1.3 Connecting observed and unobserved quantities

For $X \xrightarrow{\lambda} Y$ without confounding,

$$\sigma_{xy} = \mathbb{E}[XY] = \mathbb{E}[X(\lambda X + \mathcal{E}_y)] = \lambda \underbrace{\mathbb{E}[XX]}_{1} + \underbrace{\mathbb{E}[X\mathcal{E}_y]}_{0} = \lambda.$$

With $X \leftrightarrow Y$ (ϵ_{xy}), $X = \mathcal{E}_x$ and

$$\sigma_{xy} = \lambda \mathbb{E}[XX] + \mathbb{E}[X\mathcal{E}_y] = \lambda + \mathbb{E}[\mathcal{E}_x\mathcal{E}_y] = \lambda + \epsilon_{xy}.$$

A curious property. For $X \xrightarrow{\lambda_{xz}} Z \xrightarrow{\lambda_{zy}} Y$:

$$\begin{aligned}\sigma_{xy} &= \mathbb{E}[X(\lambda_{zy}Z + \mathcal{E}_y)] = \lambda_{zy}\mathbb{E}[XZ] + \mathbb{E}[X\mathcal{E}_y] = \lambda_{zy}\mathbb{E}[X(\lambda_{xz}X + \mathcal{E}_z)] \\ &= \lambda_{zy}\lambda_{xz}\mathbb{E}[XX] + \lambda_{zy}\mathbb{E}[X\mathcal{E}_z] = \lambda_{zy}\lambda_{xz} + \lambda_{zy}\mathbb{E}[\mathcal{E}_x\mathcal{E}_z].\end{aligned}$$

Without confounding $\sigma_{xy} = \lambda_{zy}\lambda_{xz}$; with $X \leftrightarrow Z$ (ϵ_{xz}), $\sigma_{xy} = \lambda_{zy}\lambda_{xz} + \lambda_{zy}\epsilon_{xz}$.

4.1.4 Treks and Wright's rule

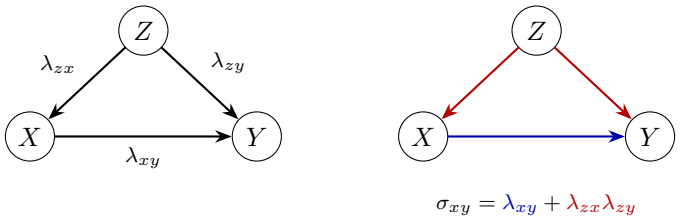


Figure. *Wright's rule: σ_{xy} sums the coefficient products over the treks, here the direct path (blue) and the trek through Z (red).*

For $Z \to X \to Y$, $Z \to Y$: $\sigma_{xy} = \lambda_{xy} + \lambda_{zx}\lambda_{zy}$, from $X \to Y$ and $X \leftarrow Z \to Y$.

Theorem 4.1 (Wright's rules, 1921)

For an acyclic linear SCM with standardized variables, σ_{xy} equals the sum, over all *open* paths between X and Y (non-self-intersecting paths without colliding arrowheads, i.e. treks; a bidirected edge counts as one step contributing ϵ), of the product of the path coefficients along the path. In particular $\sigma_{xy} = 0$ when X and Y are d-separated by $\emptyset$.

Proof [Supplement]. Write $\mathbf{V} = \Lambda^\top \mathbf{V} + \boldsymbol{\epsilon}$ (Λ strictly triangular in a topological order) and $\Omega = \mathrm{Cov}(\boldsymbol{\epsilon})$ (off-diagonal entries: bidirected coefficients). Then $\mathbf{V} = (I - \Lambda^\top)^{-1}\boldsymbol{\epsilon}$, $\Sigma = (I - \Lambda)^{-\top}\Omega(I - \Lambda)^{-1}$, and since Λ is nilpotent, $(I - \Lambda)^{-1} = \sum_{k \geq 0} \Lambda^k$ with $(\Lambda^k)_{ij}$ the sum over length-k directed paths $i \to j$ of coefficient products. Hence

$$\sigma_{xy} = \sum_{a,b} \Big(\sum_{\text{directed } a \to x} \prod \lambda \Big) \omega_{ab} \Big(\sum_{\text{directed } b \to y} \prod \lambda \Big),$$

a sum over pairs of directed paths to x and y joined at a common source $a = b$ (weight ω_{aa}) or by $a \leftrightarrow b$ (weight ω_{ab}): exactly the treks. With standardized variables, the error variance at a common-cause source combines with its other contributions to the variance 1, giving Wright's factor 1 at the top of each trek. If $\emptyset$ d-separates X, Y there are no treks, so $\sigma_{xy} = 0$. ■

So if $X \xrightarrow{\lambda} Y$, $\sigma_{xy} = \lambda +$ (other paths), and $\sigma_{xy} = \lambda$ if X, Y are d-separated in $\mathcal{G}_\lambda$ (edge removed).

Example 4.2

$W \to X$, $W \to Y$, $Z \to W$, $Z \to X$, $X \to Y$: $\sigma_{xy} = \lambda_{xy} + \lambda_{wx}\lambda_{wy} + \lambda_{zx}\lambda_{wz}\lambda_{wy}$ (paths $X \to Y$, $X \leftarrow W \to Y$, $X \leftarrow Z \to W \to Y$).

Example 4.3

$W \to X$, $W \to V \to Y$, $X \to Z \to Y$, $X \leftrightarrow Z$: $\sigma_{xy} = (\lambda_{xz} + \epsilon_{xz})\lambda_{zy} + \lambda_{wx}\lambda_{wv}\lambda_{vy}$.

4.2 Linear regression and causal effects

Motivation. What does regression compute, and when is it a structural coefficient?

4.2.1 The medical researcher

Does a new drug help? Doctors nationwide will follow the recommendation.

Step 1. data on drug dose X and biomarker (antibodies) Y.

Step 2. regression $Y = \beta X + e$ gives $\beta = 0.375$; the drug is authorized.

Step 3. given to everyone, the drug shows slope -1: it hurts, and not for lack of data. Why did regression (red line) fail, and can the causal effect (green line) be recovered?

4.2.2 The implicit assumption: no confounding

A causal reading of β presumes $X \leftarrow \mathcal{E}_x$, $Y \leftarrow \lambda_{xy}X + \mathcal{E}_y$, $\mathcal{E}_x \perp\!\!\!\perp \mathcal{E}_y$; then $\sigma_{xy} = \lambda_{xy}\mathbb{E}[XX] + \mathbb{E}[X\mathcal{E}_y] = \lambda_{xy}$.

The ground-truth model. If confounding cannot be excluded, $\mathcal{E}_x, \mathcal{E}_y$ are *correlated* ($X \leftrightarrow Y$) and regression is biased:

$$\sigma_{xy} = \lambda_{xy}\mathbb{E}[XX] + \mathbb{E}[\mathcal{E}_x\mathcal{E}_y] = \lambda_{xy} + \epsilon_{xy};$$

λ_{xy} is not identifiable, which explains 0.375 versus -1.

What does regression compute? $\mathbb{E}[(Y - \beta X)^2] = 1 - 2\beta\sigma_{xy} + \beta^2$ has derivative $-2\sigma_{xy} + 2\beta$, so $\beta = \sigma_{xy}$: *for standardized variables the regression coefficient is the covariance.*

4.2.3 Regression equation versus structural equation

- **Regression equation** $Y = \beta X + e$, $e \perp\!\!\!\perp X$ by construction; $\beta = \sigma_{xy} =: r_{yx}$. *No* causal claim.
- **Structural equation** $Y \leftarrow \lambda X + \mathcal{E}_y$ ($X \to Y$): claims $\mathbb{E}[Y \mid \mathrm{do}(X = x)] = \lambda x$, testable and falsifiable.

Confusing them is the "confusion of the century". Causal claims from regression implicitly assume $X_1 \xrightarrow{b_1} Y \xleftarrow{b_2} X_2$ with regressors unconfounded with Y; then OLS of $Y = \beta_1 X_1 + \beta_2 X_2 + e$ gives $\beta_1 = b_1$, $\beta_2 = b_2$.

Warning: make sure you have the right graph. If $X_1 \xleftarrow{b_1} Y \xrightarrow{b_2} X_2$, simple regressions give $r_{yx_1} = b_1$, $r_{yx_2} = b_2$, but multiple regression gives

$$\beta_1 = \frac{b_1(1 - b_2^2)}{1 - b_1^2 b_2^2}, \qquad \beta_2 = \frac{b_2(1 - b_1^2)}{1 - b_1^2 b_2^2}$$

(Appendix 4.6).

4.2.4 Testing conditional independence with regression

For Gaussian variables, $(X \perp\!\!\!\perp Y \mid Z)$ iff $r_{yx \cdot z} = 0$, the coefficient α of X in $Y = \alpha X + \beta Z$.

4.2.5 Single-door criterion: direct effects

Model $Z \xrightarrow{\lambda_{zx}} X \xrightarrow{\lambda_{xy}} Y$, $Z \leftrightarrow Y$ (ϵ_{zy}); regress $Y = \alpha X + \beta Z + e$:

$$\mathbb{E}[(Y - \alpha X - \beta Z)^2] = 1 - 2\alpha\sigma_{xy} - 2\beta\sigma_{yz} + 2\alpha\beta\sigma_{xz} + \alpha^2 + \beta^2,$$

$$\frac{\partial}{\partial\alpha} = 2\alpha - 2\sigma_{xy} + 2\beta\sigma_{xz}, \qquad \frac{\partial}{\partial\beta} = 2\beta - 2\sigma_{yz} + 2\alpha\sigma_{xz}.$$

With Wright ($\sigma_{xz} = \lambda_{zx}$, $\sigma_{xy} = \lambda_{xy} + \lambda_{zx}\epsilon_{zy}$, $\sigma_{yz} = \lambda_{xy}\lambda_{zx} + \epsilon_{zy}$):

$$\begin{aligned} \lambda_{xy} + \lambda_{zx}\epsilon_{zy} &= \alpha + \beta\lambda_{zx}, \\ \lambda_{xy}\lambda_{zx} + \epsilon_{zy} &= \alpha\lambda_{zx} + \beta, \end{aligned} \qquad \Longrightarrow \qquad \alpha = \lambda_{xy}, \quad \beta = \epsilon_{zy}$$

(unique solution when $\lambda_{zx}^2 \neq 1$). The coefficient of X is causal although Z is confounded with Y.

Theorem 4.2 (Single-door criterion; Pearl 2000)

Let λ be the coefficient of $X \to Y$ and $\mathcal{G}_\lambda$ the diagram with $X \to Y$ removed. λ is identifiable if there is a set $\mathbf{Z}$ such that (i) $\mathbf{Z}$ contains no descendant of Y, and (ii) $\mathbf{Z}$ d-separates X from Y in $\mathcal{G}_\lambda$. Moreover $\lambda = r_{yx\cdot\mathbf{Z}}$.

Proof [Supplement]. The coefficient of X in the regression of Y on $X, \mathbf{Z}$ is $r_{yx\cdot\mathbf{Z}}$. Write $Y = \lambda X + R$ (R: other parents and error). Then $r_{yx\cdot\mathbf{Z}} = \lambda + r_{rx\cdot\mathbf{Z}}$, and $r_{rx\cdot\mathbf{Z}} = 0$ iff R, X are uncorrelated given $\mathbf{Z}$. Treks between X and R are paths between X and Y in $\mathcal{G}_\lambda$; condition (ii) blocks them given $\mathbf{Z}$, and (i) prevents opening a path through a descendant of Y (which would involve R). So $\lambda = r_{yx\cdot\mathbf{Z}}$. ■

Example 4.4

$A \to B$, $A \to Y$, $B \to Y$: $\lambda_{ay} = r_{ya\cdot b}$, $\lambda_{by} = r_{yb\cdot a}$.

Example 4.5

$X \to Y$, $X \to W$, $Y \to W$, $Y \to Z$, $W \to Z$, $X \leftrightarrow Z$, $Y \leftrightarrow Z$: $\lambda_{xy} = r_{yx}$ (without $X \to Y$ every path meets a collider), and $\lambda_{wz} = r_{zw\cdot yx}$ ($\{Y, X\}$ blocks $W \leftarrow Y \to Z$, $W \leftarrow Y \leftrightarrow Z$, $W \leftarrow X \leftrightarrow Z$).

4.2.6 Back-door criterion: total effects

Theorem 4.3 (Back-door, linear; Pearl 2000)

The total effect of X on Y is identifiable if there is $\mathbf{Z}$ with no descendant of X that d-separates X from Y in $\mathcal{G}_{\underline{X}}$; it equals $r_{yx\cdot\mathbf{Z}}$.

Proof [Supplement]. In a linear Gaussian model $\mathbb{E}[Y \mid \mathrm{do}(x)]$ is linear in x with slope the total effect τ. By back-door adjustment (Chapter 2), $\mathbb{E}[Y \mid \mathrm{do}(x)] = \sum_{\mathbf{z}} \mathbb{E}[Y \mid x, \mathbf{z}]P(\mathbf{z})$, and $\mathbb{E}[Y \mid x, \mathbf{z}] = \alpha + r_{yx\cdot\mathbf{Z}}x + \boldsymbol{\beta}^\mathsf{T}\mathbf{z}$, so $\mathbb{E}[Y \mid \mathrm{do}(x)] = \alpha + r_{yx\cdot\mathbf{Z}}x + \boldsymbol{\beta}^\mathsf{T}\mathbb{E}\mathbf{Z}$ and $\tau = r_{yx\cdot\mathbf{Z}}$. ■

Example 4.6

$W \to X$, $W \leftrightarrow X$, $W \to Y$, $X \to Z \to Y$, $X \to Y$, $Z \leftrightarrow Y$. The total effect $\lambda_{xz}\lambda_{zy} + \lambda_{xy}$ equals $r_{yx \cdot w}$ (back-door with $\{W\}$); λ_{xy} alone is not obtained by also regressing on Z, because $Z \leftrightarrow Y$.

4.3 Algorithmic identification methods

Motivation. Identification means solving the polynomial system linking Σ to (λ, ϵ).

4.3.1 The equations of linear identification

For $Z \xrightarrow{\lambda_{zx}} X \xrightarrow{\lambda_{xy}} Y$, $Z \leftrightarrow Y$:

$$\Sigma = \underbrace{\begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}}_{\text{computable}} = \underbrace{\begin{bmatrix} 1 & \lambda_{xy} + \lambda_{zx}\epsilon_{zy} & \lambda_{zx} \\ \lambda_{xy} + \lambda_{zx}\epsilon_{zy} & 1 & \lambda_{zx}\lambda_{xy} + \epsilon_{zy} \\ \lambda_{zx} & \lambda_{zx}\lambda_{xy} + \epsilon_{zy} & 1 \end{bmatrix}}_{\text{expressions with unobservables}}.$$

With $\sigma_{xz} = \lambda_{zx}$,

$$\sigma_{xy} = \lambda_{xy} + \sigma_{xz}\epsilon_{zy}, \qquad \sigma_{zy} = \sigma_{xz}\lambda_{xy} + \epsilon_{zy}$$

is a full-rank 2×2 linear system:

$$\lambda_{xy} = \frac{\sigma_{xy} - \sigma_{xz}\sigma_{zy}}{1 - \sigma_{xz}^2}, \qquad \epsilon_{zy} = \frac{\sigma_{zy} - \sigma_{xz}\sigma_{xy}}{1 - \sigma_{xz}^2}.$$

Non-identifiability. $X \to Y$, $X \leftrightarrow Y$: one equation $\sigma_{xy} = \lambda_{xy} + \epsilon_{xy}$, two unknowns.

Finite identifiability. $X \to W$, $X \to Z$, $X \to Y$, each also $\leftrightarrow X$: six equations, six unknowns,

$$\begin{aligned} \sigma_{xw} &= \lambda_{xw} + \epsilon_{xw}, & \sigma_{wz} &= \lambda_{xw}\lambda_{xz} + \lambda_{xz}\epsilon_{xw} + \lambda_{xw}\epsilon_{xz}, \\ \sigma_{xz} &= \lambda_{xz} + \epsilon_{xz}, & \sigma_{wy} &= \lambda_{xw}\lambda_{xy} + \lambda_{xw}\epsilon_{xy} + \lambda_{xy}\epsilon_{xw}, \\ \sigma_{xy} &= \lambda_{xy} + \epsilon_{xy}, & \sigma_{zy} &= \lambda_{xz}\lambda_{xy} + \lambda_{xz}\epsilon_{xy} + \lambda_{xy}\epsilon_{xz}. \end{aligned}$$

Substitute $\epsilon = \sigma - \lambda$: $\sigma_{wz} = \lambda_{xz}\sigma_{xw} + \lambda_{xw}\sigma_{xz} - \lambda_{xw}\lambda_{xz}$ (likewise σ_{wy}, σ_{zy}); the last two give $\lambda_{xw} = (\sigma_{wy} - \lambda_{xy}\sigma_{xw})/(\sigma_{xy} - \lambda_{xy})$, $\lambda_{xz} = (\sigma_{zy} - \lambda_{xy}\sigma_{xz})/(\sigma_{xy} - \lambda_{xy})$; plugging into the first and clearing denominators,

$$0 = (\sigma_{xw}\sigma_{xz} - \sigma_{wz})\lambda_{xy}^2 + 2(\sigma_{xy}\sigma_{wz} - \sigma_{xw}\sigma_{xz}\sigma_{xy})\lambda_{xy} + (\sigma_{yw}\sigma_{xz}\sigma_{xy} + \sigma_{yz}\sigma_{xy}\sigma_{xw} - \sigma_{xy}^2\sigma_{wz} - \sigma_{yz}\sigma_{yw}),$$

$$\lambda_{xy} = \frac{-(\sigma_{xy}\sigma_{wz} - \sigma_{xw}\sigma_{xz}\sigma_{xy}) \pm \sqrt{(\sigma_{xy}\sigma_{wz} - \sigma_{xw}\sigma_{xz}\sigma_{xy})^2 - (\sigma_{xw}\sigma_{xz} - \sigma_{wz})\,c}}{\sigma_{xw}\sigma_{xz} - \sigma_{wz}},$$

c the constant term: at most two values, *finite identifiability.*

The three cases. *Identifiable*: one consistent value of λ_{xy}; *finitely identifiable*: finitely many; *not identifiable*: infinitely many.

Example 4.7

$W \to Z$, $W \to X$, $Z \to X$, $Z \leftrightarrow X$, $X \to Y$, $W \leftrightarrow Y$:

$$\begin{aligned} \sigma_{wz} &= \lambda_{wz}, & \sigma_{wy} &= (\lambda_{wx} + \lambda_{wz}\lambda_{zx})\lambda_{xy} + \epsilon_{wy}, \\ \sigma_{wx} &= \lambda_{wx} + \lambda_{wz}\lambda_{zx}, & \sigma_{zy} &= (\lambda_{zx} + \epsilon_{zx} + \lambda_{wz}\lambda_{wx})\lambda_{xy} + \lambda_{wz}\epsilon_{wy}, \\ \sigma_{zx} &= \lambda_{zx} + \epsilon_{zx} + \lambda_{wz}\lambda_{wx}, & \sigma_{xy} &= (\lambda_{wz}\lambda_{zx} + \lambda_{wx})\epsilon_{wy} + \lambda_{xy}. \end{aligned}$$

Here $\lambda_{xy} = (\sigma_{zy} - \sigma_{wz}\sigma_{wy})/(\sigma_{zx} - \sigma_{wz}\sigma_{wx})$ (Z is an IV conditional on W). ID algorithms find solvable subsystems efficiently; generic computer algebra is doubly exponential.

4.4 From instrumental variables to instrumental sets

Motivation. Variables affecting Y only through X yield solvable linear subsystems.

4.4.1 Instrumental variables

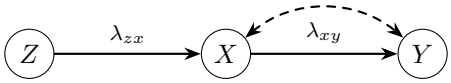


Figure. *The IV graph:* $\sigma_{zy} = \lambda_{zx}\lambda_{xy}$ *and* $\sigma_{zx} = \lambda_{zx}$ *give* $\lambda_{xy} = \sigma_{zy}/\sigma_{zx}$ *despite* $X \leftrightarrow Y$.

Add an independent doctor's recommendation Z: $Z \xrightarrow{\lambda_{zx}} X \xrightarrow{\lambda_{xy}} Y$, $X \leftrightarrow Y$. (Non-parametrically $P(y \mid \mathrm{do}(x))$ is *not* identifiable; linearity helps.)

Definition (Instrumental variable; Pearl 2000, p. 248)

Z is an IV for λ_{xy} if Z is d-separated from Y in $\mathcal{G}_{\lambda_{xy}}$ and Z is not d-separated from X in $\mathcal{G}_{\lambda_{xy}}$.

By Wright, $\sigma_{zx} = \lambda_{zx}$, $\sigma_{zy} = \lambda_{zx}\lambda_{xy}$, so

$$\lambda_{xy} = \frac{\sigma_{zy}}{\sigma_{zx}} = \frac{r_{zy}}{r_{zx}}.$$

Two-stage least squares. $X = \alpha Z + e$ gives $\alpha = \sigma_{zx}$; regressing Y on $\hat{X} = \sigma_{zx}Z$ gives $\beta = \sigma_{zy}/\sigma_{zx}$ ($\mathrm{Cov}(Y, \hat{X}) = \sigma_{zx}\sigma_{zy}$, $\mathrm{Var}(\hat{X}) = \sigma_{zx}^2$).

4.4.2 Conditional instrumental variables

Definition (Conditional IV; Brito and Pearl 2002)

Z is a conditional IV given $\mathbf{W}$ for λ_{xy} if (i) $\mathbf{W}$ contains only non-descendants of Y;
(ii) $\mathbf{W}$ d-separates Z from Y in $\mathcal{G}_{\lambda_{xy}}$;
(iii) $\mathbf{W}$ does not d-separate Z from X in $\mathcal{G}_{\lambda_{xy}}$.

Then $\lambda_{xy} = r_{yz \cdot \mathbf{W}}/r_{xz \cdot \mathbf{W}}$. Example: $W \to Z \to X \to Y$, $X \leftrightarrow Y$, $W \leftrightarrow Y$: Z is no IV ($Z \leftarrow W \leftrightarrow Y$ is open) but a conditional IV given W.

4.4.3 Instrumental sets

$Z_i \to X_j$ $(i, j \in \{1, 2\})$, $X_j \to Y$, $X_j \leftrightarrow Y$. No single IV exists (every path from Z_i to Y passes through both X_j). With $\sigma_{z_i x_j} = \lambda_{z_i x_j}$,

$$\sigma_{z_1 y} = \sigma_{z_1 x_1}\lambda_{x_1 y} + \sigma_{z_1 x_2}\lambda_{x_2 y}, \qquad \sigma_{z_2 y} = \sigma_{z_2 x_1}\lambda_{x_1 y} + \sigma_{z_2 x_2}\lambda_{x_2 y},$$

solvable by Cramer's rule if the matrix is non-singular.

When does the system fail? If $Z_1 \to Z_2 \to \{X_1, X_2\}$, paths from Z_1 to Y are those from Z_2 times $\lambda_{z_1 z_2}$:

$$\lambda_{z_1 z_2}\sigma_{z_2 y} = \lambda_{z_1 z_2}(\sigma_{z_2 x_1}\lambda_{x_1 y} + \sigma_{z_2 x_2}\lambda_{x_2 y}), \qquad \sigma_{z_2 y} = \sigma_{z_2 x_1}\lambda_{x_1 y} + \sigma_{z_2 x_2}\lambda_{x_2 y},$$

proportional equations: degenerate.

Theorem 4.4 (Non-intersecting paths)

The system is non-degenerate (generically) iff there is a full matching between $\{Z_1, Z_2\}$ and $\{X_1, X_2\}$ by vertex-disjoint (non-intersecting) paths.

Proof [Supplement]. The equations $\sigma_{z_i y} = \sum_j \sigma_{z_i x_j}\lambda_{x_j y}$ have a unique solution iff $\det[\sigma_{z_i x_j}] \neq 0$. Each $\sigma_{z_i x_j}$ is a trek sum, and by Lindström–Gessel–Viennot the determinant is a signed sum over vertex-disjoint path systems $\{Z_1, Z_2\} \to \{X_1, X_2\}$: none $\Rightarrow$ all terms cancel, determinant $\equiv 0$; one $\Rightarrow$ a nonzero polynomial, nonzero for generic parameters. ($k \times k$: Brito and Pearl 2002.) ■

In the first graph $Z_1 \to X_1$, $Z_2 \to X_2$ are disjoint; in the degenerate one both paths pass through Z_2. Max-flow finds such paths: sources Z_i, sinks X_j, unit vertex capacities; flow $|\mathbf{Z}|$ certifies a full matching.

4.4.4 Conclusions

1. Linear SCMs are the most common causal models in practice.
2. Know what regression actually computes.
3. A structural equation is not a regression equation, though regression sometimes recovers structural coefficients.
4. The diagram can be leveraged to solve for causal effects in general settings.

4.5 Appendix: Complexity of identification in linear SCMs

Landscape. Single-door, IV and instrumental sets are sufficient but incomplete; the half-trek criterion and auxiliary cutsets/ACID (Chen et al. 2016, Kumor et al. 2019) cover more, not all, identifiable instances.

Limits. Graphical criteria are polynomial but incomplete; Gröbner bases are complete but doubly exponential (a 4-node graph can take over 75 days; García-Puente, Spielvogel and Sullivant 2010; Bardet 2002).

Recent result (Dörfler, van der Zander, Bläser and Liśkiewicz, NeurIPS 2024). The general problem is harder than NP (real-valued unknowns) but lies in PSPACE, hence $2^{\text{poly}(n)}$ time instead of $2^{2^{\text{poly}(n)}}$. Use graphical criteria when they work; otherwise the problem is hard but solvable.

4.6 Appendix: Multiple regression when Y is the cause

$X_1 \xleftarrow{b_1} Y \xrightarrow{b_2} X_2$: $\sigma_{yx_1} = b_1$, $\sigma_{yx_2} = b_2$, $\sigma_{x_1x_2} = b_1b_2$. The normal equations of $Y = \beta_1 X_1 + \beta_2 X_2 + e$ are

$$b_1 = \beta_1 + \beta_2 b_1 b_2, \qquad b_2 = \beta_2 + \beta_1 b_1 b_2.$$

Substituting $\beta_2 = b_2 - \beta_1 b_1 b_2$: $b_1 = \beta_1 + b_1 b_2^2 - \beta_1 b_1^2 b_2^2$, so

$$\beta_1 = \frac{b_1(1 - b_2^2)}{1 - b_1^2 b_2^2}, \qquad \beta_2 = \frac{b_2(1 - b_1^2)}{1 - b_1^2 b_2^2}.$$

4.7 Appendix: The Gröbner basis approach

Identification as polynomial elimination. Covariances are polynomials in λ, ϵ. For $X_1 \xrightarrow{\lambda_{12}} X_2 \xrightarrow{\lambda_{23}} X_3$, $X_2 \leftrightarrow X_3$ (ϵ_{23}):

$$\sigma_{12} = \lambda_{12}, \qquad \sigma_{13} = \lambda_{12}\lambda_{23}, \qquad \sigma_{23} = \lambda_{23} + \epsilon_{23},$$

so $\lambda_{23} = \sigma_{13}/\sigma_{12}$ (X_1 is an instrument), while σ_{23} alone cannot separate λ_{23} from ϵ_{23}. Gröbner bases automate this elimination.

How it works.

1. *Set up:* $\sigma_{ij} -$ (polynomial in λ, ϵ) $= 0$ for every pair (i, j).
2. *Eliminate:* elimination term orders keep only the target λ_{xy} and the σ_{ij}.
3. *Check:* degree 1 in λ_{xy}: generically identifiable; degree d: algebraically d-identifiable (d possible values); λ_{xy} absent: not generically identifiable.

Computational results (García-Puente et al. 2010; all linear SCMs on 3 and 4 nodes):

	3 nodes ($2^6 = 64$)	4 nodes ($2^{12} = 4096$)
Generically identifiable	22	1246
Not generically identifiable	42	2844
Algebraically k-identified	0	6
Solved by single-door / IV	22/22	1093/1246
Only by Gröbner basis	0	153

On 4 nodes, 153 cases need the algebraic method; runtimes range from seconds to over 75 days per graph.